



BAULKHAM HILLS HIGH SCHOOL

JULY 2016
YEAR 11 TASK 2

Mathematics

General Instructions

- Reading time 5 minutes
- Working time – 60 minutes
- Write using black or blue pen
- Board-approved calculators may be used
- Show all necessary working in questions 6 to 9
- Marks may be deducted for careless or badly arranged work

Total marks – 53

Exam consists of 6 pages.

This paper consists of TWO sections.

Section 1 – Page 2-3

Questions 1-5 (5 marks)

- Attempt Question 1-5

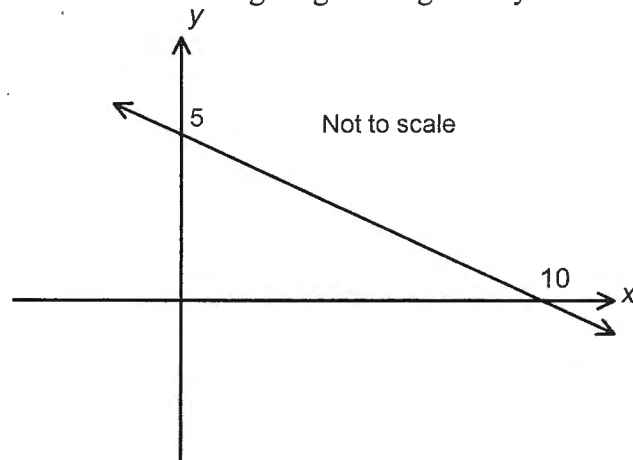
Section II – Pages 3-6 (48 marks)

- Attempt questions 6 to 9

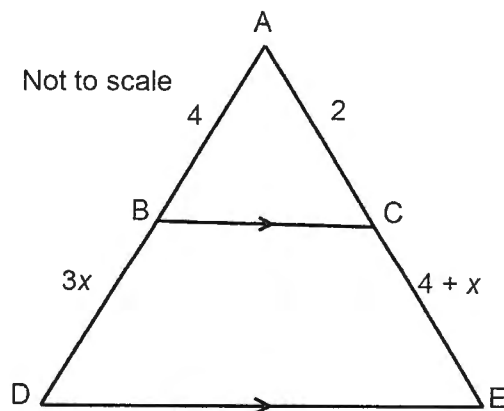
Section I - 5 marks
Attempt questions 1-5

Use the multiple choice answer page for question 1-5

1. The equation of the line in the following diagram is given by



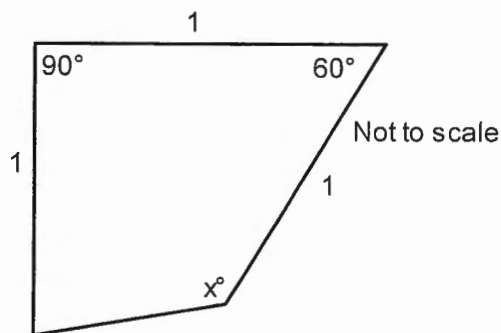
- (A) $y = \frac{1}{2}x + 5$ (B) $y = 2x + 5$
 (C) $y = -\frac{1}{2}x + 5$ (D) $y = -\frac{1}{2}x + 10$
2. The length of AD in the following is:



- (A) 1.6 (B) 8 (C) 14 (D) 28
3. The derivative of $\frac{1}{(x^2+1)^2}$ is:

- (A) $\frac{-2}{(x^2+1)}$ (B) $\frac{-4x}{(x^2+1)}$ (C) $\frac{-2}{(x^2+1)^3}$ (D) $\frac{-4x}{(x^2+1)^3}$

4. In the diagram find the value of x .



- (A) 120
(B) 135
(C) 137.5
(D) 140

5. Evaluate $\lim_{x \rightarrow \infty} \frac{4x^2}{9 - 6x + 7x^2}$

- (A) 0
(B) $\frac{4}{9}$
(C) $\frac{4}{7}$
(D) 1

Section 2 Questions 6-9

6. (a) Differentiate :

(i) $6x^2 - 4x - 3$

1

(ii) $\sqrt[3]{x} - \frac{4}{x^3}$

2

(iii) $\frac{2x}{3x^2 - 4}$

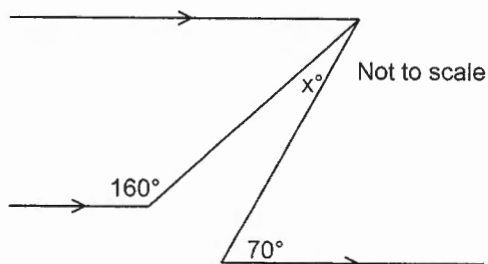
2

(iv) $(4x^2 - 3)^6$

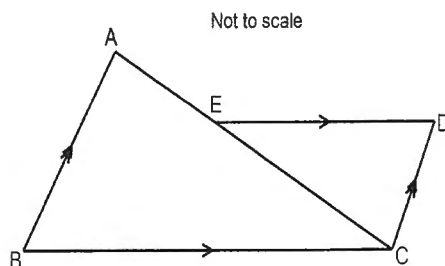
2

- (b) Find the value of the pronumeral (no reasons required)

1



c)

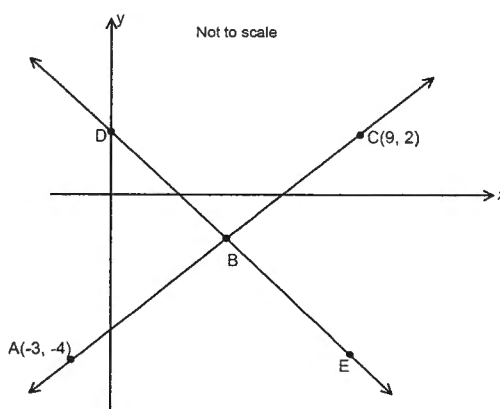


In the above diagram $AB \parallel CD$ and $DE \parallel BC$.

(i) Prove that $\triangle ABC \sim \triangle EDC$. 2

(ii) If $3CD = 2AB$ and $DE + BC = 20$ cm, find the length of DE. 3

7. (a) The points $A(-3, -4)$ and $C(9, 2)$ are drawn on the number plane below. B is the midpoint of AC and BD is perpendicular to AC.



(i) Find the exact distance AC. 1

(ii) Find the co-ordinates of B. 1

(iii) Find the gradient of AC. 1

(iv) Show the equation of BD is $2x + y - 5 = 0$. 2

(v) What are the co-ordinates of the point D where BD cuts the y axis? 1

(vi) Find the point E on BD such that $\triangle ABD \equiv \triangle BCE$. 1

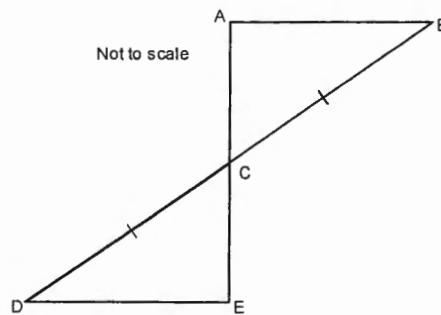
(vii) What type of quadrilateral is AECD? Justify your answer. 2

(b) Differentiate $x^2(6 - x)^5$. 3

8. (a) Find the equation of the normal to the curve $y = 2 - 3x^2$ at the point where $x = 2$ 3
 Answer in general form.

- (b) The perpendicular distance from the point $(3, k)$ to the line $2x + 4y + 5 = 0$ is $3\sqrt{5}$. Find the value(s) of k . 3

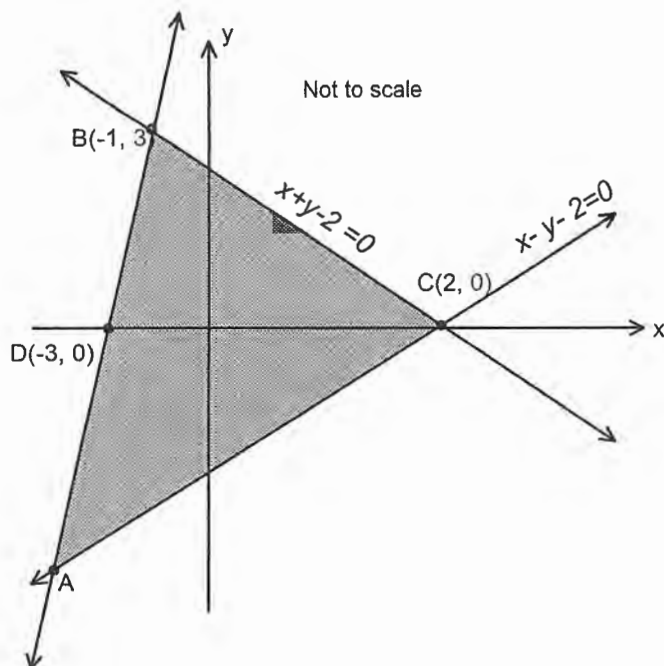
(c)



In the above diagram $BC = CD$, $AE \perp AB$ and $AE \perp DE$.

- (i) Prove $\triangle ABC \equiv \triangle CDE$. 3
- (ii) Hence prove ABED is a parallelogram. 2
- (iii) If $AB = 6$ cm and $BD = 18$ cm find the area of the parallelogram ABED. 2

9.



(a) The diagram above shows a region enclosed by the lines $x - y - 2 = 0$, $x + y - 2 = 0$ and the line AB.

- | | | |
|-------|--|---|
| (i) | Show that the equation of the line AB is $3x - 2y + 9 = 0$. | 1 |
| (ii) | Show that the co-ordinates of point A are (-13, -15). | 2 |
| (iii) | Give a set of inequalities that define the shaded region. | 2 |
| (iv) | Find the area of the shaded region | 2 |

(b)	If $y = \frac{2x+1}{\sqrt{4x+2}}$ show that $\frac{dy}{dx} = \frac{1}{2y}$.	3
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~ End of Exam ~

Solutions
Multiple Choice
 1. C, 2. D 3. D, 4. B, 5. C

6a) (i) $y' = 12x - 4$ ①
 (ii) $y = x^{\frac{1}{3}} - 4x^{-3}$ ①
 $y' = \frac{1}{3}x^{-\frac{2}{3}} + 12x^{-4}$ ①
 (iii) $\frac{d}{dx} \left(\frac{2x}{3x^2-4} \right) =$
 $\frac{2(3x^2-4) - 6x(2x)}{(3x^2-4)^2}$ ①
 $= \frac{6x^2 - 8 - 12x^2}{(3x^2-4)^2}$
 $= \frac{-6x^2 - 8}{(3x^2-4)^2}$ ①
 (iv) $\frac{d}{dx} \left[(4x^2-3)^6 \right]$ ①
 $= 48x(4x^2-3)^5$ ①
 b) $x = 50^\circ$ ①

c) (i) $\angle BAC = \angle ACD$ (Alternate $\angle$'s) ①
 $\angle ACB = \angle DEC$ (Alternate $\angle$'s) ①
 $DE \parallel BC$
 $\therefore \triangle ABC \parallel \triangle DEC$ (AA) ①
 (ii) $3CD = 2AB$
 $\therefore \frac{CD}{AB} = \frac{2}{3} \therefore \frac{DE}{BC} = \frac{2}{3}$
 (matching sides in the same ratio in $\triangle$'s) ①
 let $DE = x \therefore BC = 20 - x$
 $\therefore \frac{2}{3} = \frac{x}{20-x}$ ①
 $2(20-x) = 3x$
 $40 - 2x = 3x$
 $5x = 40$
 $x = 8$
 $\therefore DE = 8\text{cm}$ ①
 7a) (i) $AC = \sqrt{12^2 + 6^2}$
 $= \sqrt{180}$ ①
 $= (6\sqrt{5})$ ①
 (ii) $B \left(-\frac{3+9}{2}, -\frac{4+2}{2} \right)$ ①
 $= (3, -1)$ ①
 (iii) $m_{AC} = \frac{2+4}{9+3} = \frac{1}{2}$ ①

(iv) $m_{BD} = -2$ ①
 $\therefore y - y_1 = -2(x - x_1)$
 $y + 1 = -2x + 6$ ①
 $2x + y - 5 = 0$
 (v) when $x = 0, y = 5$ D(0,5) ①
 (vi) $\frac{0+x}{2} = 3$ $\frac{5+y}{2} = -1$ ①
 $x = 6$ $y = -7$
 $E(6, -7)$
 (vii) $DE = \sqrt{12^2 + 6^2}$
 $= 6\sqrt{5} = AC$ ①
 $\therefore DAEC$ is a square
 diagonals $=, \perp$ & bisect each other. ①
 b) $\frac{d}{dx} (x^2(6-x)^5)$ ①
 $= 2x(6-x)^5 + 5x^2(6-x)^4(-1)$ ①
 $= 2x(6-x)^5 - 5x^2(6-x)^4$ ①
 $= x(6-x)^4 [2(6-x) - 5x]$ ①
 $= x(6-x)^4 (12 - 7x)$ } not necessary
 ① mark for applying product rule
 ① " for derivative of $6-x$
 ① " for the final answer.

8a) $y = 2 - 3x^2$
 at $x = 2, y = -10$ ①
 $y' = -6x$
 at $x = 2, y' = -12 \rightarrow m = -12$
 $\perp m = \frac{1}{12}$ ①
 $\therefore$ Normal
 $y + 10 = \frac{1}{12}(x - 2)$
 $12y + 120 = x - 2$ ①
 $x - 12y - 122 = 0$
 b) $\frac{|6 + 4k + 5|}{\sqrt{2^2 + 4^2}} = 3\sqrt{5}$ ①
 $\frac{|11 + 4k|}{\sqrt{20}} = 3\sqrt{5}$ ①
 $|11 + 4k| = 30$
 $11 + 4k = \pm 30$

$11 + 4k = 30 \rightarrow k = \frac{19}{4}$
 $11 + 4k = -30 \rightarrow k = \frac{-41}{4}$
 $\therefore k = \frac{19}{4}, \frac{-41}{4}$
 c) (i) $BC = ED$ (Given)
 $\angle DCE = \angle ACB$ (Vertically Opposite $\angle$'s) ①
 $\angle CAB = 90^\circ$ ($AE \perp AB$)
 $\angle CED = 90^\circ$ ($AE \perp DE$)
 $\therefore \angle CED = \angle CAB$ ①
 $\triangle ABC = \triangle CAB$ ①
 $\triangle ABC = \triangle CDE$ (AAS) ①
 (ii) $AC = CE$ (Matching sides in $\triangle$'s)

$\therefore ABED$ is a parallelogram
 as Diagonals BD & AE bisect each other.
 (iii) $BC = 9\text{cm}$
 $\therefore AC = \sqrt{9^2 - 6^2}$
 $= 3\sqrt{5}$ ①
 Area = $AB \times AE$
 $= 6 \times 2(3\sqrt{5})$
 $= 36\sqrt{5}$ ①
 (80.5cm^2)

$$9 \text{ (i) } m_{AB} = \frac{3-0}{-1+3} \\ = \frac{3}{2}$$

$$\therefore \text{BD } y-0 = \frac{3}{2}(x+3) \quad (1)$$

$$2y = 3x+9$$

$$3x-2y+9=0$$

(ii) can sub $(-13, -15)$ into both lines or solve simult'ly.

$$x-y-2=0 \quad \text{--- (1)}$$

$$3x-2y+9=0 \quad \text{--- (2)}$$

$$\text{from (1) } y = x-2$$

sub in (2)

$$3x-2(x-2)+9=0 \quad (1)$$

$$3x-2x+4+9=0$$

$$x = -13$$

$$y = -13-2 \\ = -15$$

$$A(-13, -15) \quad (1)$$

$$(ii) \quad x-y-2 \leq 0$$

$$x+y-2 \leq 0$$

$$3x-2y+9 \geq 0$$

(1) if 1 inequality is correct

+ (1) - all correct.

$$(iv) \text{ Area} = \Delta BDC + \Delta DAC$$

$$= \frac{1}{2} \times 5 \times 3 + \frac{1}{2} \times 5 \times 15$$

$$= 45 \text{ units} \quad (1)$$

Alternatively could do area using $\perp$ distance

$$b) \quad y = \frac{2x+1}{\sqrt{4x+2}}$$

$$y' = \frac{2(\sqrt{4x+2}) - \frac{1}{2}(4x+2)^{-\frac{1}{2}} \cdot 4(2x+1)}{(\sqrt{4x+2})^2} \quad (1)$$

$$= \frac{2\sqrt{4x+2} - \frac{2(2x+1)}{\sqrt{4x+2}}}{(4x+2)}$$

$$(4x+2)$$

$$= \frac{2(4x+2) - 2(2x+1)}{\sqrt{4x+2} \cdot 4x+2}$$

$$= \frac{8x+4-4x-2}{(\sqrt{4x+2})(4x+2)}$$

$$= \frac{4x+2}{(\sqrt{4x+2})(4x+2)}$$

$$= \frac{1}{\sqrt{4x+2}} \quad (1)$$

now

$$\frac{1}{2y} = 1 \div \frac{2(2x+1)}{\sqrt{4x+2}}$$

$$= \frac{1 \times \sqrt{4x+2}}{(4x+2)} \quad (1)$$

$$= \frac{1}{\sqrt{4x+2}}$$

$$= \frac{dy}{dx}$$

$$\therefore \frac{dy}{dx} = \frac{1}{2y}$$